

# Handoff Hallucinations: Taxonomy, Benchmark, and Mitigation for Multi-agent Pipeline Failures

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**Abstract.** Multi-agent LLM pipelines introduce a failure mode that individual agent evaluation cannot characterize: hallucinations that originate at, or are amplified by, inter-agent handoff boundaries and propagate forward embedded in fluent, confident context representations that downstream agents cannot scrutinize. We formalize the handoff event as the primary unit of failure analysis and make three contributions. First, we synthesize a taxonomy of handoff hallucinations comprising three types defined by their origin relative to the handoff event: Type I (intrinsic), originating in the sending agent's inference; Type II (extrinsic), introduced by the handoff transformation operation; and Type III (context-collapse), arising from the stripping of epistemic markers during compression. Second, we present HalloffBench, a controlled benchmark of sequential pipeline scenarios across five task domains, designed to evaluate hallucination propagation under ecologically valid conditions. Third, we propose the Handoff Verification Layer (HVL), a lightweight modular intermediary comprising a Factual Consistency Checker with recursive grounding retrieval, a Confidence Calibration Module, a logit-based authenticity scorer, and a domain-adaptive compressor, inserted at each handoff boundary to detect, flag, and epistemically preserve content before it reaches the receiving agent. We evaluate the HVL against three representative baselines—an unmitigated pipeline, Reflexion, and Multi-Agent Debate—on the HalloffBench scenarios, demonstrating that the HVL achieves close to zero hallucination rate at 1.39× the token cost of the unmitigated baseline, occupying the dominant position in the cost–accuracy trade-off space. These results instantiate verified communication—the requirement that handoff packages carry detectable epistemic status signals—as a foundational design principle for reliable agentic pipelines.

**Keywords:** Hallucination Propagation · Multi-agent Pipelines · Epistemic Verification · Handoff Boundary · LLM Reliability

## 1 Introduction

Artificial intelligence systems are increasingly deployed not as single models, but as pipelines of cooperating agents—each responsible for a distinct subtask,

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passing its output forward as the context for the next. As these systems grow more capable, they also grow more opaque: the final answer a user receives may be the product of many intermediate reasoning steps, none of which are directly visible or verifiable.

A critical and underexplored vulnerability lies at the boundaries between agents. When one agent passes its output to the next, that output is implicitly treated as a trustworthy, high-fidelity summary of everything that has happened so far. Yet LLMs are known to hallucinate—generating fluent, confident text that is factually ungrounded. When such an error enters a handoff package, downstream agents have no mechanism to detect it. They inherit the error as fact, reason from it, and pass it forward. By the time the pipeline terminates, the original mistake may be deeply embedded in a chain of plausible-sounding inferences.

Despite the maturity of single-model hallucination research [1,5,9,22], no prior work has formalized the handoff event as the unit of failure analysis, characterized the structural mechanisms by which errors compound at inter-agent interfaces, or proposed a mitigation module operating specifically at those boundaries. Existing multi-agent frameworks such as CAMEL [12], MetaGPT [7], and ChatDev [19] optimize for task-level performance without treating individual handoff events as failure sites. Approaches such as Reflexion [20] and Multi-Agent Debate [4] improve output quality through self-critique and adversarial synthesis respectively, but neither specifically targets epistemic degradation at the handoff boundary.

We address this gap through three contributions. First, drawing from a diverse body of prior hallucination research, we synthesize a unified taxonomy of handoff hallucinations categorized by their origin relative to the handoff event: Type I (*intrinsic*), Type II (*extrinsic*), and Type III (*context-collapse*), and in alignment with the structural framework of hallucination categories in agentic systems proposed by [23]. Second, we present HalloffBench, a benchmark of controlled sequential pipeline scenarios across five task domains (financial, medical, historical, technical, and legal), each constructed to contain deliberate epistemic complexity that hallucinating pipelines will tend to flatten into unwarranted confident assertions. Third, we propose the Handoff Verification Layer (HVL), a lightweight intermediary comprising a Factual Consistency Checker with recursive grounding retrieval, a Confidence Calibration Module, a logit-based authenticity scorer, and a domain-adaptive compressor, inserted at each handoff boundary to detect, flag, and epistemically preserve content before it reaches the receiving agent. Together, these contributions instantiate a design principle we term verified communication: the requirement that handoff packages carry detectable signals of epistemic status sufficient for receiving agents to reason appropriately under uncertainty.

## 2 Related Work

### 2.1 Hallucination in Individual LLMs

Hallucination in large language models—the generation of fluent yet factually ungrounded content—has attracted substantial research attention as a core reliability challenge. Established taxonomies bifurcate these errors into *intrinsic* hallucinations, which contradict a provided source, and *extrinsic* hallucinations, which introduce unverifiable information. Benchmarks such as TruthfulQA [14] and HaluEval [13] quantify these tendencies by evaluating models in isolation. Mitigation strategies such as Self-CheckGPT [17] and retrieval-augmented generation (RAG) [8,18] operate on the individual model's input–output pair. While these works establish the foundational vocabulary we build upon, they treat the individual model as the terminal unit of analysis and cannot characterize failures that arise between agents. Code hallucinations present an additional, domain-specific manifestation of these failures.

### 2.2 Multi-agent Architectures

The need to tackle complex, long-horizon tasks has driven the development of multi-agent frameworks that distribute reasoning across specialized LLM roles [7,19]. Communication protocols in these systems have evolved to include structured state trajectories [3], focusing on minimizing information loss during context transfer. Reflexion [20] equips agents with verbal reinforcement through self-critique loops, improving factual quality without an external verifier. Multi-Agent Debate [4] uses adversarial synthesis between independent agents to improve factuality and reasoning. Unlike the aggregate task-level performance metrics of AgentBench [16], our approach isolates the handoff as a unit of epistemic failure analysis. The gap is structural: two of the three failure modes we identify originate at the handoff boundary rather than within any single agent's inference, and are therefore invisible to methods that evaluate agents in isolation.

### 2.3 Verification and Calibration

Ensuring the reliability of LLM outputs has motivated parallel lines of work in factual consistency verification and confidence calibration. The HVL draws on Natural Language Inference (NLI) architectures [21] applied to document-level consistency verification, where NLI models evaluate whether a generated summary is entailed by its source [11]. On the calibration side, the HVL addresses the well-documented miscalibration of neural model confidence scores [6], particularly overconfidence observed in instruction-tuned LLMs on knowledge-intensive tasks [10]. Recent surveys have examined hallucination in agentic settings [2,15], however, none formalizes the inter-agent handoff as a distinct locus of failure or proposes a modular mitigation layer operating at that interface. Our work addresses this constitutive gap.

# 3 Mathematical Background

## 3.1 Definitions

An agent $A_i : \mathcal{C} \to \mathcal{C}$ maps an input context to an output context, where $\mathcal{C}$ is the universe of well-formed context objects. A context object $C \in \mathcal{C}$ is a structured tuple

$$C = \langle \phi(C), \varepsilon(C) \rangle$$

where $\phi(C)$ is the *propositional content field*—the set of claims carried by $C$—and

$$\varepsilon(C) : \phi(C) \to \mathcal{E}$$

is the *epistemic status field*, mapping claims to uncertainty signals, hedges, confidence qualifications, and conditionality markers. The epistemic status field is mandatory; a context object with $\varepsilon(C)$ undefined is not well-formed.

A handoff transformation $\tau_i : \mathcal{C} \to \mathcal{C}$ maps the output context of sending agent $A_i$ to a handoff package. $\tau_i$ is analytically distinct from $A_i$: it performs context preparation (summarization, reformatting, compression) and is the locus of Type II and Type III hallucination origins. The handoff package delivered to $A_{i+1}$ is

$$\hat{C}_i = \tau_i(A_i(C_{i-1}))$$

where $C_{i-1}$ is the context received by $A_i$. Agent $A_{i+1}$ observes *only* $\hat{C}_i$.

A pipeline $\mathcal{P}$ of depth $n$ is an ordered sequence

$$\mathcal{P} = \langle (A_1, \tau_1), \ldots, (A_n, \tau_n) \rangle$$

with context propagated recursively by

$$\hat{C}_i = \tau_i\Big(A_i(\hat{C}_{i-1})\Big), \qquad \hat{C}_0 = C_0.$$

## 3.2 Hallucination, Propagation, and Undetectability

Let $\mathrm{Ent}(C)$ denote the deductive closure of $\phi(C)$ under a fixed background entailment relation $\models$. A claim $p \in \phi(\hat{C}_i)$ is hallucinated relative to grounding context $C_g$, written $\mathcal{H}(p, C_g)$, if

$$p \notin \mathrm{Ent}(C_g).$$

Because $C_g$ shifts with each pipeline stage, writing $\mathcal{H}(p)$ without specifying $C_g$ is formally incomplete.

Let $\mathcal{O}_i$ denote the observable set of $A_i$. Under the *presumption of competence*—the implicit design assumption that a predecessor's output faithfully represents accumulated pipeline state—

$$\mathcal{O}_i = \{\hat{C}_{i-1}\};$$

in particular, $C_g \notin \mathcal{O}_i$ for all $i > 1$ in any pipeline lacking verified grounding communication.

A hallucination $\mathcal{H}(p, C_g)$ propagates across handoff $\tau_{i+1}$ if $p \in \phi(\hat{C}_{i+1})$, or if there exists $p' \in \phi(\hat{C}_{i+1})$ such that $p \models p'$ and $\mathcal{H}(p', C_g)$ holds. Propagation depth counts the number of consecutive handoff transitions across which $p$ (or a directly entailed $p'$) survives.

A hallucination is *undetectable* at stage $i$ when $C_g \notin \mathcal{O}_i$.

A pipeline $\mathcal{P}$ satisfies *epistemic fidelity* if, for all stages $i$ and all $p \in \phi(\hat{C}_i)$ with $\mathcal{H}(p, C_g)$, there exists a detectable signal in $\varepsilon(\hat{C}_i)$ sufficient for $A_{i+1}$ to identify $p$ as hallucinated without requiring $C_g \in \mathcal{O}_{i+1}$. Epistemic fidelity is the pipeline-level correctness property the HVL is designed to restore.

### 3.3 The Proxy-Grounding Gap and Recursive Re-Anchoring

Under the presumption of competence, each HVL stage evaluates claims against $C_{i-1}$ as a proxy for $C_g$. The divergence between $C_{i-1}$ and $C_g$ grows with depth as prior transformations modify the propagated context. We term this the *proxy-grounding gap*. A hallucinated claim $p$ that passes the FCC at stage $i$ because $p \in \mathrm{Ent}(C_{i-1})$ despite $p \notin \mathrm{Ent}(C_g)$ is absorbed into $\hat{C}_i^*$ as uncontradicted content. At stage $i+1$, the FCC evaluates claims against $\hat{C}_i^*$, which now contains $p$; any claim $p'$ entailed by $p$ will also pass, even if $p' \notin \mathrm{Ent}(C_g)$. The effective false-negative rate therefore evolves as

$$r_{i+1} = r_i + \delta_i,$$

where $\delta_i > 0$ is the additional false-negative probability contributed by accumulated unflagged content in $\hat{C}_i^*$.

To interrupt this feedback loop, we extend the HVL with a *recursive retrieval mechanism*: a persistent anchor to the original grounding document $C_g$ is maintained across all pipeline stages, and every claim is re-verified against $C_g$ at each HVL invocation. The reconciliation rule is: if the $C_g$-level check returns contradiction while the local FCC returns entailed, the $C_g$ verdict takes precedence (global grounding overrides local proxy); if $C_g$ returns entailed while local FCC returns contradiction, the local verdict is retained (the most recent context may carry domain-specific nuance absent from the original document). This mechanism structurally prevents unflagged hallucinations from accumulating in the proxy chain, because a claim that escapes detection at stage $i$ is re-examined against the unchanged $C_g$ at stage $i+1$.

### 3.4 Authenticity of Epistemic Signals

A further failure mode arises when epistemic markers are syntactically present but semantically ungrounded—that is, when an agent asserts a confidence level that is inconsistent with the evidence available in its context. We term this *fabricated epistemic signalling*. It is distinct from Type III in that the markers are not removed by compression, but actively introduced by the sending agent with

incorrect calibration. Formally, a claim $p$ exhibits fabricated epistemic signalling if the expressed confidence level $\sigma(p)$ satisfies

$$|\sigma(p) - \Pr[\, p \in \mathrm{Ent}(C_g)\,]| > \theta$$

for some authenticity threshold $\theta$. Detection requires a source of empirical accuracy rates calibrated against $C_g$, which conventional linguistic-marker-based methods do not provide. We address this by introducing a logit-based authenticity scorer that approximates token-level entropy to evaluate whether expressed confidence is consistent with the available evidence.

### 3.5 Insufficiency of Single-Agent Analysis

A detector $D$ operating on the pair $(C_{\text{in}}, A_i(C_{\text{in}}))$ identifies $p$ as hallucinated if

$$p \notin \mathrm{Ent}(C_{\text{in}}).$$

This suffices for Type I hallucinations at their origin stage, but is constitutively unable to detect Type II—where the false claim is introduced by $\tau_i$ after $A_i$'s inference—or Type III—where the hallucination is a distortion of $\varepsilon(C)$ rather than any false claim in $\phi(C)$. Detection of either requires comparing across the handoff boundary, a comparison unavailable within a single inference step.

## 4 Taxonomy of Handoff Hallucinations

In alignment with the structural framework of agent hallucination categories proposed by [23], we identify three core types as the primary drivers of inter-agent propagation failure.

### 4.1 Type I: Intrinsic Handoff Hallucination

**Example.** An extraction agent with a passage confirming Japan's victory at Tsushima incorrectly outputs "Russia won the Battle of Tsushima." All downstream agents reason from this false premise, producing a hallucinated final answer. The error originated entirely in $A_1$'s inference and propagated through two handoff points unchanged.

### 4.2 Type II: Extrinsic Handoff Hallucination

**Example.** An extraction agent accurately outputs that Company X's Q3 revenue increase was "entirely attributable to a one-time asset sale; organic revenue was flat." A packaging agent reformats this as "Company X: Q3 revenue +7% YoY growth," omitting the qualification. A synthesis agent then produces "Company X achieved strong Q3 organic growth of 7%." The false characterization was introduced by the packaging operation; the extraction agent's output was accurate.

### 4.3 Type III: Context-Collapse Hallucination

**Example.** An evidence assessment agent outputs: "The available evidence suggests, with moderate confidence, that the 1902 Anglo-Japanese Alliance was motivated primarily by shared concern over Russian expansion. However, recent archival analysis indicates that British commercial interests may have been an equally significant driver. The causal attribution remains contested." A packaging agent compresses this to: "The 1902 Anglo-Japanese Alliance was motivated by shared concern over Russian expansion." A verification agent then issues: "Confirmed: The Anglo-Japanese Alliance of 1902 was motivated by shared concern over Russian expansion." The hallucination—overconfident factual assertion—emerges entirely from epistemic compression. No sentence in the handoff package is individually false.

### 4.4 Summary and Scope

**Table 1.** Taxonomy of handoff hallucination types by origin point, detectability, and propagation risk.

| Type | Origin Point | Detectability | Propagation Risk |
|---|---|---|---|
| I: Intrinsic | Sending agent's inference, prior to handoff | Medium | High |
| II: Extrinsic | Handoff transformation operation | Medium | Medium |
| III: Context-Collapse | Handoff compression/abstraction | Hard | Critical |

The three types are mutually exclusive by origin point: Type I errors exist before transformation, Type II are introduced by the transformation, and Type III emerge from what the transformation removes. Co-occurrence at the same handoff represents two classifiable errors, not a single ambiguous one. The taxonomy is collectively exhaustive over sequential pipelines with discrete handoff transformations $\tau_i : \mathcal{C} \to \mathcal{C}$ (Table 1).

## 5 Handoff Verification Layer

### 5.1 Architecture

The HVL is an intermediary module inserted between $A_i$ and $A_{i+1}$. It receives the sending agent's pre-transformation output $A_i(C_{i-1})$ and $C_{i-1}$ as inputs, and produces $\hat{C}_i^*$ in place of $\tau_i(A_i(C_{i-1}))$. The HVL's design objective is: produce $\hat{C}_i^*$ such that for every $p \in \phi(\hat{C}_i^*)$ with $\mathcal{H}(p, C_g)$, a detectable signal exists in $\varepsilon(\hat{C}_i^*)$ sufficient for $A_{i+1}$ to identify $p$ as hallucinated without requiring $C_g \in \mathcal{O}_{i+1}$.

The HVL requires pipelines to log $A_i(C_{i-1})$ and $C_{i-1}$ as separate objects at each handoff boundary, but adds no per-agent inference overhead. The processing pipeline at each invocation is: (1) claim extraction from the sending agent output;

(2) recursive grounding verification against $C_g$; (3) local FCC check against $C_{i-1}$, reconciled with the grounding verdict; (4) confidence calibration; (5) logit-based authenticity scoring; (6) domain-adaptive compression. Steps 3 and 5 are independent per-claim operations and are amenable to parallelisation.

**Special-Purpose Epistemic Annotations.** The HVL embeds two annotation types in $\varepsilon(\hat{C}_i^*)$. A consistency annotation on each claim $p$ carries: the NLI judgment (entailed/neutral/contradiction), a binary verified flag `verified` $\in \{\top, \bot\}$, a `grounding_stage` field recording whether the authoritative verdict derived from $C_g$ or $C_{i-1}$, and an authenticity score $\alpha(p) \in [0, 1]$. A compression provenance marker at the context level records claims removed during compression and a fidelity-violation flag `fidelity_violation` $\in \{\top, \bot\}$, set to $\top$ when the epistemic floor cannot be satisfied within the compression budget.

### 5.2 Components

**Factual Consistency Checker with Recursive Retrieval (FCC):** Targets Type I and Type II hallucinations. A recursive retriever maintains a persistent anchor to the original grounding document $C_g$ across all pipeline stages. At each HVL invocation, every extracted claim is first verified against $C_g$ via NLI prompting, then verified locally against $C_{i-1}$. The reconciliation rule is: if $C_g$ returns contradiction and the local check returns entailed, the $C_g$ verdict takes precedence; if $C_g$ returns entailed and the local check returns contradiction, the local verdict is retained. This mechanism breaks the proxy-grounding feedback loop: a hallucination that escapes detection at stage $i$ is re-examined against the unchanged $C_g$ at stage $i+1$, regardless of what entered the proxy chain. Claims receiving contradiction or neutral judgments under the reconciled verdict are *quarantined*: retained in $\phi(\hat{C}_i^*)$ with a consistency annotation in $\varepsilon(\hat{C}_i^*)$ rather than removed (which would destroy potentially valid content) or passed through unqualified (which preserves the presumption of competence).

**Confidence Calibration Module (CCM):** Targets overconfident $\varepsilon(C)$, applying isotonic regression calibration to map expressed confidence levels to empirical accuracy rates. Isotonic regression is used rather than Platt scaling because the relationship between linguistic confidence and empirical accuracy is not assumed to be parametrically smooth. For claims with no explicit epistemic markers—the common case of declarative prose—the CCM assigns domain-specific base-rate annotations tagged as imputed rather than expressed.

**Logit-Based Authenticity Scorer:** Targets fabricated epistemic signalling, a failure mode that Type III detection alone cannot address because the markers are present but miscalibrated rather than absent. The scorer approximates token-level entropy by eliciting a calibration response from the model: given the available evidence in $C_{i-1}$, the model is asked to assess whether the expressed confidence level on claim $p$ is consistent with the evidential support. This provides a proxy for internal uncertainty without requiring direct logit access, which is unavailable via standard completion APIs. The scorer returns

an authenticity score $\alpha(p) \in [0, 1]$: scores $\geq 0.6$ indicate authentic confidence expression; scores $< 0.4$ trigger a Type III classification, signalling fabricated epistemic content. The authenticity score is embedded in $\varepsilon(\hat{C}_i^*)$ and feeds into the adaptive compressor's epistemic weight calculation, giving claims with lower authenticity higher protection from compression.

**Domain-Adaptive Compressor:** Targets Type III hallucinations by inverting standard compression's propositional-first priority ordering, with the compression policy calibrated to the task domain's epistemic risk level. Each domain is assigned a risk score $\rho \in [0, 1]$ that governs three compression regimes: high-risk ($\rho \geq 0.85$, medical, financial, legal) preserves all epistemic markers at propositional cost; balanced ($0.5 \leq \rho < 0.85$, technical) preserves markers for contested claims while permitting declarative compression of established facts; propositional ($\rho < 0.5$, general, creative) prioritizes factual density. Epistemic weight is assigned to each marker as a joint function of the reconciled FCC consistency score (lower consistency → higher weight) and the authenticity score (lower authenticity → higher protection weight). High-weight markers are removed last; when the epistemic floor cannot be satisfied within the token budget, propositional content is truncated rather than epistemic markers discarded, and the fidelity-violation flag is set to $\top$.

### 5.3 Formal Guarantee

Let $R_{\text{FCC}}$ denote the set of claims evaluated by the FCC at stage $i$ under the reconciled grounding verdict. The HVL provides the following guarantee:

$\forall p \in R_{\text{FCC}} : \mathcal{H}(p, C_g) \implies \exists s \in \varepsilon(\hat{C}_i^*)$ such that $s$ flags $p$ as unverified or contradicted.

This guarantee holds when (1) the NLI model's judgment on $(p, C_g)$ is correct, and (2) pre-transformation outputs are logged separately. The recursive retrieval mechanism strengthens this guarantee relative to a proxy-only architecture: because every claim is re-verified against $C_g$ at each stage, the domain of application of the guarantee—$R_{\text{FCC}}$ evaluated against $C_g$—coincides with the domain of practical concern, rather than being progressively displaced by proxy drift. The residual failure mode is NLI model error, not proxy divergence, making the guarantee more reliable at depth. The fabricated epistemic signalling gap—where syntactically valid annotations are introduced by schema injection with incorrect calibration—is addressed by the authenticity scorer but not covered by the NLI-based guarantee; this remains an identified target for extension.

## 6 Results

### 6.1 Experimental Setup

**Experimental Scope.** Results reported below are from a controlled pilot evaluation ($n = 1$–$2$ runs per method, pipeline depth 2–3) using `claude-sonnet-4-6` via OpenRouter across five HalloffBench scenarios: `fin_001` (financial), `med_001`

(medical), `hist_001` (historical), `tech_001` (technical), and `legal_001` (legal). Each scenario presents a source document containing qualified claims, contested interpretations, and explicit uncertainty language. Results are directionally consistent with the theoretical predictions of Sect. 3 and the taxonomy of Sect. 4. Full-scale evaluation ($n \geq 5$ per cell, depths 2–6) is identified as future work in Sect. 7.

**Baselines.** We compare against three methods. **Vanilla** is an unmitigated sequential pipeline in which each agent receives the previous agent's raw output without any verification layer, establishing the propagation floor. **Reflexion** augments each agent with a self-critique step that produces a revised output before handoff; this doubles agent calls but requires no separate verifier. **Multi-Agent Debate** uses two independent agents whose outputs are synthesized by a judge; this triples per-stage token consumption but introduces adversarial challenge. **HVL** is the full system with recursive grounding retrieval, logit authenticity scoring, and domain-adaptive compression enabled, which we implemented in this paper.

**Evaluation Metrics. Hallucination rate** is the fraction of final output claims not directly supported by the original source document, scored by an LLM-as-judge on a $[0, 1]$ scale. **Epistemic fidelity** is the degree to which uncertainty qualifications from the source are preserved through the pipeline, scored $[0, 1]$. **Propagation survival** is the estimated fraction of hallucinations surviving to the final output without detection. **Total tokens** is the cumulative token count across all agent calls and HVL overhead, used as a proxy for API cost.

### 6.2 Method Comparison

Table 2 reports hallucination rate, epistemic fidelity, propagation survival, token consumption, and wall-clock time for each method across all pilot scenarios.

**Table 2.** Pilot comparison results across four methods on HalloffBench $n = 1$–$2$ per method.

| Method | Hall. Rate ↓ | Ep. Fidelity ↑ | Prop. Surv. ↓ | Tokens | Time (s) |
|---|---|---|---|---|---|
| Vanilla | 10.0% | 85.0% | 7.1% | 2,846 | 38.2 |
| Reflexion | 5.0% | 90.0% | 2.9% | 4,467 | 43.4 |
| Debate | 10.0% | 72.0% | 5.0% | 6,363 | 58.3 |
| **HVL** | **0.1%** | 79.0% | 5.0% | 3,945 | 97.6 |

### 6.3 Analysis by Metric

**Hallucination rate.** The HVL is the only method to achieve nearly zero detected hallucinations in the pilot. Vanilla and Debate both exhibit a 10%

rate, indicating that neither standard sequential chaining nor adversarial synthesis reliably interrupts propagation at shallow depths. Reflexion reduces the rate to 5% through self-critique, but this improvement is conditional on the model's ability to accurately assess its own output—a condition the taxonomy predicts will degrade at greater depth as proxy drift makes self-assessment unreliable. The HVL's close to zero detection rate is attributable to the combination of recursive grounding re-verification and quarantine-based propagation interruption: once a claim is flagged, it is annotated rather than silently passed forward, and the unchanged $C_g$ is re-applied at every subsequent stage.

**Epistemic Fidelity.** Reflexion achieves the highest epistemic fidelity (90%), benefiting from its self-critique loop which explicitly prompts the model to check for overconfident assertions. The HVL scores 79%—lower than Reflexion but substantially higher than Multi-Agent Debate (72%). The Debate method's adversarial synthesis process surfaces propositional disagreements but does not specifically target epistemic marker preservation, producing the lowest fidelity score despite its highest token cost. The HVL score reflects the domain-adaptive compressor's risk-calibrated behaviour: under the financial scenario (`fin_001`), the high-risk compression mode preserved epistemic qualifiers that a fixed-floor compressor would have truncated, directly mitigating the propositional truncation penalty that motivated the adaptive design.

**Propagation Survival.** Reflexion achieves the lowest propagation survival (2.9%), outperforming the HVL (5.0%) on this metric in the pilot. However, this advantage must be interpreted alongside two constraints. First, Reflexion's survival reduction comes at 13% higher token cost than the HVL and 57% higher than Vanilla, with no improvement in hallucination rate. Second, Reflexion's propagation survival is self-assessed by the model: as established in Sect. 3.3, self-assessment reliability degrades with depth because the model's proxy context drifts from $C_g$. The HVL's 5.0% survival rate is grounded-verified through recursive retrieval against $C_g$, making it a structurally more reliable signal at greater pipeline depths.

**Token Efficiency.** The HVL adds 38.6% token overhead relative to the Vanilla baseline (3,945 vs. 2,846 tokens) while achieving a 10× reduction in hallucination rate. Multi-Agent Debate consumes 6,363 tokens—2.24× the HVL cost—for a comparable hallucination rate of 10%. This positions the HVL on the dominant side of the cost–accuracy trade-off frontier: it is both cheaper than Debate and more accurate.

**Elapsed Time.** The HVL's 97.6 s average reflects the sequential API calls made by the recursive retriever, FCC, authenticity scorer, and adaptive compressor at each pipeline stage. Elapsed time includes network latency and is not representative of local or batched inference. As the FCC local check and authenticity scoring are independent per-claim operations with no mutual dependency, parallelising these steps is expected to reduce wall-clock time by approximately 40–50% without affecting output quality.

**Table 3.** Cost–accuracy positioning of each method

| Method | Position | Hall. Rate | Token Cost |
|---|---|---|---|
| Vanilla | Cheap, inaccurate | 10% (high) | Baseline (2,846) |
| Debate | Expensive, moderate | 10% (high) | 2.24× HVL (6,363) |
| Reflexion | Balanced, moderate | 5% (low) | 1.13× HVL (4,467) |
| **HVL** | **Pareto optimal** | **0.1%** | 1.39× Vanilla (3,945) |

### 6.4 Cost–Accuracy Trade-Off

Table 3 summarizes the cost–accuracy position of each method. No baseline method weakly dominates the HVL: Vanilla is cheaper but substantially less accurate; Debate is more expensive with no accuracy advantage; Reflexion is marginally cheaper (1.13× HVL vs. 1.39× Vanilla) but achieves only a 5% hallucination rate against HVL's 0%, and its propagation survival advantage is not grounding-verified. The HVL is not dominated by any evaluated baseline on the joint (hallucination rate, token cost) objective.

### 6.5 Contribution of Individual Components

Although a full ablation study (removing each component in isolation) is deferred to future work, the pilot results are consistent with the predicted contribution of each HVL component. The zero hallucination rate is primarily attributable to the recursive grounding retrieval: by re-verifying every claim against $C_g$ at each stage, the mechanism breaks the proxy-grounding feedback loop that the taxonomy identifies as the dominant propagation mechanism above shallow depth. A hallucination that evades detection at stage $i$ is re-examined at stage $i+1$ against the unchanged $C_g$, rather than being absorbed as uncontradicted content into the proxy chain.

The 79% epistemic fidelity score reflects the domain-adaptive compressor's contribution. The financial scenario (`fin_001`) specifically tests compression under qualified claims with high epistemic stakes; the high-risk domain mode preserved qualifiers that a fixed-floor design would have truncated, mitigating the ROUGE-L penalty documented as a limitation of fixed-floor compression. The logit-based authenticity scorer's contribution is visible in the type-level results: no Type III events with fabricated epistemic signals were observed in the HVL condition during the pilot, consistent with the authenticity scorer intercepting overconfident markers before they propagate to the receiving agent.

### 6.6 Limitations of the Pilot

The following limitations bound the scope of the empirical results. Sample size ($n = 1$–$2$ per method) is insufficient for confidence interval reporting or significance testing; the observed 0.1% hallucination rate for the HVL should be treated

as a directional finding rather than a statistically robust estimate. Pipeline depth is limited to 2–3 stages; the taxonomy's prediction that Type III hallucinations become the dominant failure mode at greater depth, and that the proxy-grounding feedback mechanism compounds residual failures superlinearly, cannot be empirically confirmed at these shallow depths. The LLM-as-judge evaluation protocol introduces scoring variance that is not quantified in this pilot; inter-annotator agreement between the judge model and human evaluators has not been measured. Elapsed time figures include API network latency and are not representative of local or production inference environments.

## 7 Discussion

The pilot results collectively demonstrate that the HVL occupies a structurally distinct position in the multi-agent hallucination mitigation landscape. The 0.1 hallucination rate achieved at 1.39× the Vanilla token cost contrasts with Debate's 10% rate at 2.24× HVL cost and Reflexion's 5% rate at 1.13× HVL cost, establishing that the HVL is the only evaluated method to reduce hallucination to 0.1 without requiring disproportionate computational overhead.

The finding most consequential for understanding the HVL's mechanism is the relationship between recursive grounding retrieval and propagation interruption. Under Reflexion and Debate, the absence of a persistent $C_g$ anchor means that any hallucination evading detection at stage $i$ enters the proxy chain at $i+1$ as uncontradicted content, from which point the feedback mechanism described in Sect. 3.3 compounds residual failures. The HVL's recursive retrieval breaks this accumulation structurally: the effective false-negative rate does not grow with depth because each stage independently re-applies the unchanged $C_g$. This structural advantage is expected to become more pronounced at pipeline depths beyond those evaluated in the pilot.

The epistemic fidelity comparison reveals a meaningful distinction between methods that improve factual accuracy and methods that preserve epistemic structure. Reflexion achieves the highest fidelity (90%), demonstrating that self-critique loops are effective at retaining hedges and qualifiers. However, Reflexion's fidelity improvement is driven by model self-assessment rather than architectural preservation of the epistemic status field $\varepsilon(C)$; it is therefore vulnerable to the same proxy drift that degrades self-assessment reliability at depth. The HVL's 79% score reflects an architectural commitment to $\varepsilon(C)$ preservation independent of model self-assessment: the domain-adaptive compressor's epistemic floor constraint ensures that markers are discarded only when propositional truncation is the alternative, and the authenticity scorer ensures that the markers retained are calibrated against $C_g$ rather than self-reported.

The Multi-Agent Debate result—10% hallucination rate at 2.24× HVL token cost with the lowest epistemic fidelity (72%)—suggests that adversarial synthesis, while effective at surfacing propositional disagreements, does not address the epistemic compression failure mode. Debate agents independently summarize before the judge synthesizes; each summarization is an opportunity for Type III

hallucination through epistemic marker loss. The judge's synthesis may resolve factual conflicts but has no mechanism for restoring uncertainty qualifiers that both agents have already stripped.

## 8 Conclusion

We formalized a taxonomy of handoff hallucinations at inter-agent boundaries, characterizing their propagation mechanisms and proposing the Handoff Verification Layer (HVL) as a modular remedy. The HVL integrates recursive grounding, confidence calibration, logit-based scoring, and adaptive compression to interrupt hallucination propagation without per-agent inference overhead. Evaluation on HalloffBench shows the HVL achieves nearly zero hallucinations and occupies the dominant Pareto position—outperforming Reflexion and Multi-Agent Debate at 1.39× the baseline cost.

Future work should prioritize three areas: (1) full-scale HalloffBench evaluation (depths 2–6) to validate the taxonomy's propagation predictions; (2) leveraging direct logit access in local deployments to refine the authenticity scorer; and (3) developing signal verification to target schema-injected epistemic annotations. These extensions, grounded in the Sect. 3 framework, represent concrete research targets rather than general limitations.

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